

SIDDHARTH SHARMA

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OBJECTIVE

B.Tech CSE student at IIIT Kottayam with hands-on AI/ML research and software development experience. Skilled in Python, Django, FastAPI, TensorFlow, Keras, scikit-learn, NumPy, and Pandas, with experience developing and evaluating lightweight ML models and building backend applications. Interested in applying AI/ML to real-world software systems.

EDUCATION

Indian Institute of Information Technology (IIIT), Kottayam	2024 – 2028
B.Tech in Computer Science and Engineering	CGPA: 7.69/10

RESEARCH & PUBLICATIONS

Research Intern – ABV-IIITM Gwalior	<i>Full-Time May 2026 – Jul 2026</i>
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- Developed a lightweight TinyML intrusion detection pipeline on **2.2M+ network-flow records** using EDA, preprocessing, feature engineering, Random Forest feature selection, and autoencoder-based anomaly detection.
- Designed Dense and Undercomplete Autoencoders using Python, TensorFlow/Keras, scikit-learn, NumPy, and Pandas for resource-constrained IoT environments.
- Reduced input features from **63 to 20** while achieving **99.78% accuracy** with the Undercomplete Autoencoder and reducing model parameters by **35% (956 to 624)**.
- Reduced model size from **43.37 KB to 39.47 KB** and inference latency from **0.17 ms to 0.06 ms/sample**. Evaluated models across 845,090 test samples, with the Dense Autoencoder achieving **99.85% F1-score and 100% recall**.

Publication – ICIIS 2026: “*TinyML-Based Intrusion Detection: A Comparative Study of Autoencoders for Smart City IoT Devices*” – **Accepted**

TECHNICAL SKILLS

Languages	Python, C/C++, Java, JavaScript, TypeScript, SQL
AI/ML	Machine Learning, Neural Networks, TensorFlow, scikit-learn, TinyML, Anomaly Detection
Data Science	NumPy, Pandas, EDA, Data Preprocessing, Feature Engineering, Model Evaluation
Web/Backend	Django, FastAPI, REST APIs, React.js, Next.js, HTML5, CSS3, Tailwind CSS
Databases/Tools	PostgreSQL, MySQL, MongoDB, Git, GitHub, Linux, Postman, Vercel

EXPERIENCE

Full Stack Developer Intern – Decodelabs	<i>Remote May 2026 – Jun 2026</i>
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- Built **StudyHub**, a full-stack productivity application with study-session tracking and analytics.
- Built REST APIs using **FastAPI/Python**, integrated **PostgreSQL** end-to-end, and developed the responsive frontend using HTML5/CSS3.

PROJECTS

Khatu Shyam Trading Co. – E-Commerce Platform	<i>TypeScript, Next.js, Tailwind CSS, PostgreSQL</i>
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- Developing a B2B e-commerce platform with product search, filtering, sorting, trade pricing, inventory status, product details, minimum-order quantities, cart, and customer accounts.
- Implemented responsive product browsing with WhatsApp order inquiries, contact forms, Google Maps integration, and support for multiple brands and product categories.

ACHIEVEMENTS & LEADERSHIP

- JEE Main 2024:** 96.46 percentile (Top 4% nationally).
- Lead, Chitrachaya – IIIT Kottayam:** Led 15+ members across 10+ events and mentored 8 recruits.
- Gold Medal** — MP Tourism Quiz | **1st Place** — Inter-School Debate.
- Certifications:** Google Cloud Gen AI Academy; Advances in Robotics, IIIT Kottayam.